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IOT Based Automatic Pet Feeder

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ABSTRACT

We own pets for their companionship gives us emotional support. It helps to reduce our stress and sense of loneliness. We treat pets as part of our families. We always want to take good care of our pets supplying timely nutritious food. Often times we may not be able to supply food timely to our pets. The Internet of Things (IoT) technology can improve quality of life by intelligently connecting physical devices through internet. We here investigate the application of IoT to automate the process of pet feeding. We use two feeders one for solid food and one for liquid food. The dispenser of solid food is controlled by a DC servo motor and that of liquid food is controlled by a DC solenoid valve. The motor and solenoid valve will be controlled by ATSAMD21 chip microcontroller. A ublox Wi-Fi module is used along with the microcontroller to enable the connection of actuators to the internet. A camera module is also used so that the owner of the pet can monitor remotely the pet's activities. The owner of the pet can control the pet feeders from anywhere in the world and also can monitor his pets, through any internet enabled device like smart phone. The process of pet feeding can also be automated by the owner by configuring the microcontroller appropriately.

I. INTRODUCTION

IMPORTANCE OF PETS

Pets are part of our everyday lives and part our families. They provide as with companionship but also with emotional support, reduce our stress levels, sense of loneliness and helps us to increase our social activities and add to a child's self- esteem and positive emotional development.

PETS ARE GOOD FOR HEALTH

If you spend around 15 minutes petting your favourite cat or dog, your body will release the following natural "feel good" hormones: oxytocin, prolactin and serotonin. It also lowers your cortisol, which is the body's natural stress hormone. Not only will this send your body into a relaxed state, but it can also lower your blood pressure by 10% too. If you own a dog, then you will have to walk it at least once or twice a day which means

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you will be getting vital fresh air and exercise while having the opportunity to connect with nature and bond with your four- legged friend.

PETS ARE GOOD FOR KIDS

Research shows that children who grow up in homes with pets are likely to have a stronger immune system and less likely to develop childhood allergies and asthma. Research carried out by health psychologist Dr June Mc Nicholas of the University of Warwick and drug company Novartis Animal Health tested the saliva of 138 children aged between four and 11-years-old for IgA antibodies- an antibody found in the immune system that helps fight off infection. Results showed that pet- owning children were able to fend off infection such as colds and flu better than those children without pets, thus having less days off school.

IMPACT OF PETS ON ENVIRONMENT

The biggest environmental impact associated with our animal companions comes from producing meat based pet food, which uses land, water and energy resources and is a significant source of greenhouse gas emissions. Dogs and cats both eat meat-heavy diets, meaning they contribute to the consumption of animal products, the meat farming industry and all the pollutions, toxins and other emissions that it creates. Since our pets need the nutrients in animal products in order to have a healthy diet, finding a way to minimize their impact on the environment proves challenging.

SUSTAINABLE

Until the pet food industry catches up with American's demands for more sustainable options, pet owners can follow these tips to be eco-friendlier:

- Adjust your pet's diet with guidance from a veterinarian: If you have found a more sustainable pet food option that you'd like to try under the supervision of your veterinarian. You never know how your dog will react to a change in diet, so it's always best to be proactive.
- Use biodegradable poop bags: If you're using old plastic grocery bags for pet poop clean-up, consider switching to biodegradable poop bags. These are much better for the environment and also smaller, so you aren't wasting a whole grocery bag and while you're at it, switch to reusable grocery bags.

PET FEEDING

METHODS OF PET FEEDING

Overconsumption of food is a leading cause of pet obesity. Food is measured using a standard cooking cup to weigh the amount of food your veterinarian recommends for your pet's daily intake. Feed your pet at the same time each day to regulate digestive system and avoid accident.

PROVIDING ENOUGH WATER

Pets should have access to clean, fresh water at all times. If you are concerned with your pet is not consuming enough water, feeding food helps encourage water consumption

FREQUENCY OF PET FEEDING

Puppies should be fed 4-6 times a day. This should gradually decrease by the time the pet reaches 50percent of his adult body wait. At this stage pet should be fed 3meals per day. Smaller, more frequent feedings might be suitable for pets with health problems that decrease their appetite.

can see your pet and talk to them. You can also snap photos of your pet through the Treat Cam. The video quality is spot on with this smart feeder. There's also a social media network called Petzi for users to share images of their pets. I'm not sure thatwill take off, but it's a good marketing tool.

Wagz Smart Dog Feeder

Wagz Smart Dog Feeder may be the perfect fit for larger dogs. It has a large capacity to hold plenty of food for large dogs or multiple dogs. You can feed your dog from anywhere through your smartphone and the Wagz app. There's also audio capabilities that allow you to talk to your dog. Through the mobile app, you can control portion sizes based on details of your dog, like height, weight, and age. There's also an option to place automatic food orders through Amazon dash when the feeder runs low. You must sign up for the service to take advantage of this program.

Pet net Smart Feeder

Get alerts when your pet has been fed as well as if there were any issues during a feeding. You'll even get a notification when it's time to clean the pet feeder.

Take control of your pet's health with specially designed portion sizes determined through the app.

II. EXISTING METHODS

AUTOMATED PET FEEDER

Paola A. Buitrago – Cp. MalcolmA. Morgan – Cp. E, Hector L. Rodriguez – E.E, proposed the content analysis of Automated Pet Feeder include a collection of requirements that are meant to ease the burden of pet ownership. One objective is that the Automated Pet Feeder must continually monitor the weight of the food and water supply of both its dispensers, while also holding a minimum quantity of food and water capable of feeding a large dog for an extended period of time, particularly several days. The Automated Pet Feeder will dispense food and water in portions of weight and volume, respectively, that will have been specified by the pet's owner, and these portions must be able to be specified from an application on the pet owner's mobile device. Data and general information regarding the dog's eating patterns will be sent to the pet owner's mobile device, also including the ability to stream live footage of the apparatus's environment from a camera located on the apparatus. The ability to differentiate between an authorized pet and unauthorized pets will be accomplished using collar recognition, which is one ofthe primary features that sets apart the Automated Pet Feeder from its competition.

A Remote Pet Feeder Control System via MQTT Protocol

Wen-Chuan Wu1, Ke-Chung Cheng1, and Pei-Yu Lin2, proposed the content analysis of A Remote Pet feeder control systems via MQTT Protocol owner can control the feeder to throw food at their pet in a remote mode through a smartphone. Some feeders also have a camera function that allows the owner to observe the pet at home through the camera. However, these machines are stationary and can hardly move. The pictures are also in a fixed shooting angle, yet pets may move around the house. Therefore, this paper will design a remote control system on a toy car that is equipped with an IP camera, feeding stuffs and water. The proposed remote control system is not only a pet monitor but also considered as a pet feeder. Compared with the existing smart pet feeders, it is movable and interactive, which can enhance the interaction between the owner andthe pet.

Design of Pet Feeder using Web Server as Internet of ThingsApplication Andi Adriansyah, Muchd. Arief Wibowo, Eko Ihsanto proposed the content analysis of Pet Feeder as a tool to pet the animals which is equipped with a communication network and can be controlled over the distance through a web server, it is different from the pet feeder tool in general, it works when the ownergave orders through a web server. The system applied to the Pet Feeder worked well, the time to display the web server only takes for about four seconds, while to determine the availability of food in containers and tanks could be done by utilizing some software and the internet access even the controller was in the distant places. After designing and testing the tool, it can be concluded that the web server will show some data such as the distance of the food container detected by ultrasonic sensor, the bit value generated by LDR sensor, the servo motor position, and the link to feed the animals. Servo motors can be operated in two methods, i.e. using a web server and push button. The time required to open a web server is around 4 seconds, but the weakness of the tool is web server cannot be accessed if the ESP8266 module does not get a public IP.

Automatic Pet Food Dispenser by using Internet of Things (IoT)

Tannop Sangvanloy and Kingkarn Sookhanaphibarn proposed the content analysis of Automatic Pet Food Dispenser by using Internet of Things (IoT). The objective of this project is to develop an automatic pet feeding with Internet of Things (IoT). We developed a feeder to help allocate dry food diet to small pets such as dogs and cats. The objective of this proposed machine is to make a convenience for the pets'owners and make a promotion to raise their pets in healthy fashion. To raise a healthy, the pet feeding for each meal has to be seriously considered. When feeding is not timely, it will directly affect the health of the pet. It can cause malnutrition and various illnesses or diseases. Therefore, this proposed machine will be a solution for the above-mentioned problem by controlling the feeding operation through a mobile application.

III. PROPSED SYSTEM

In our proposed system we had done the project by using automatic pet feeder by adding some components to overcome the existing method. The heart of the component is Arduino MKR Wi-Fi 1010 and the pin 5V is

connected to the relay at VCC and the pin 6 is connected to the relay IN1 and pin GND is connected to the relay GND. The relay NO1 is connected to the solenoid valve and the COM1 is connected to battery negative. The solenoid valve is connected to the battery positive. Then the solenoid valve is connected to the liquid food dispenser. The Arduino MKR Wi-Fi 1010 the pin GND is connected to the servo motor negative and the pin s at the servo motor. Then the servo motor is connected to the solid food dispenser. Here the Arduino board act as web server. In which the web server has an IP address. When we enter the IP address in the web page. By using these IP address, we have login to the web page. We enter these IP address in any web page the Arduino board is connected to internet. In this webpage we have two buttons ON and OFF. Here we have Arduino board as a web server. It has an IP address. When we enter the IP address in web browser a new web page is open. The webpage has two buttons ON and OFF. Along with this Arduino board we interfaced the solenoid valve and servo motor. Solenoid valve is used to feed solid food. The solid food is kept in a container we use servo motor for closing and opening of the lead. The liquefied things are kept in the bottle in which the bottle is placed downward direction. The solenoid valve is attached with it through relay. When the solenoid valve is activated the liquid flows otherwise the liquid does not flow. The solenoid valve and servo meter is controlled by the web page in which it has ON and OFF button. If the button is ON, the solenoid valve and servo motor is ON. If the webpage button is OFF, the solenoid valve and servo motor is also OFF. Solenoid valve is controlled by the relay. The signal is give through the solenoid valve is open. The relay has two mode of operation,

- Normally closed
- Normally open

If the relay is in normally open the signal does not flow, so the value is in OFF condition. If the relay is normally closed the signal flows to the value, then we can operate the value for the feeding liquefied things to the pet. We use this method to feed the pets in a healthy manner.

IV. BLOCK DIAGRAM

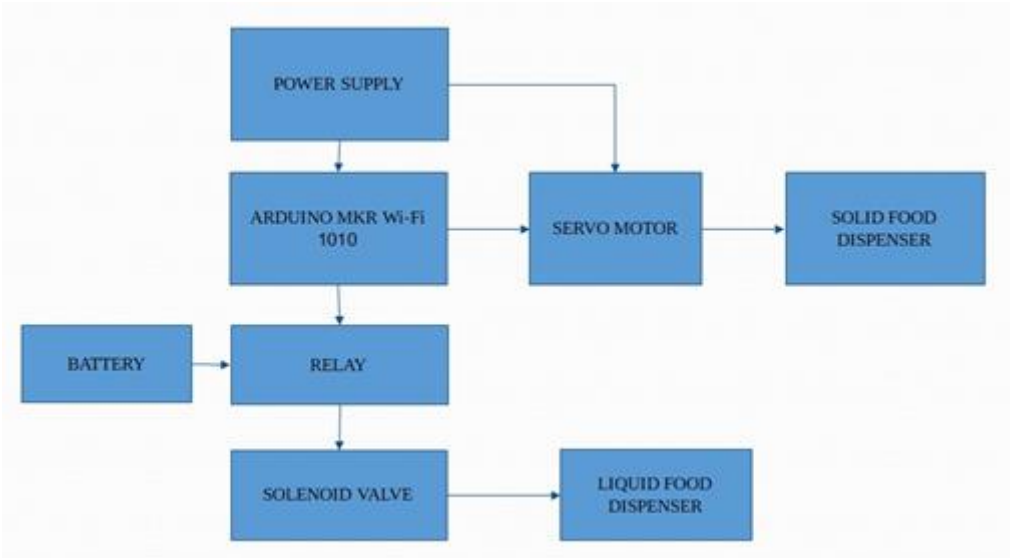

Fig4.1: Block diagram of proposed system

V. CIRCUIT DIAGRAM

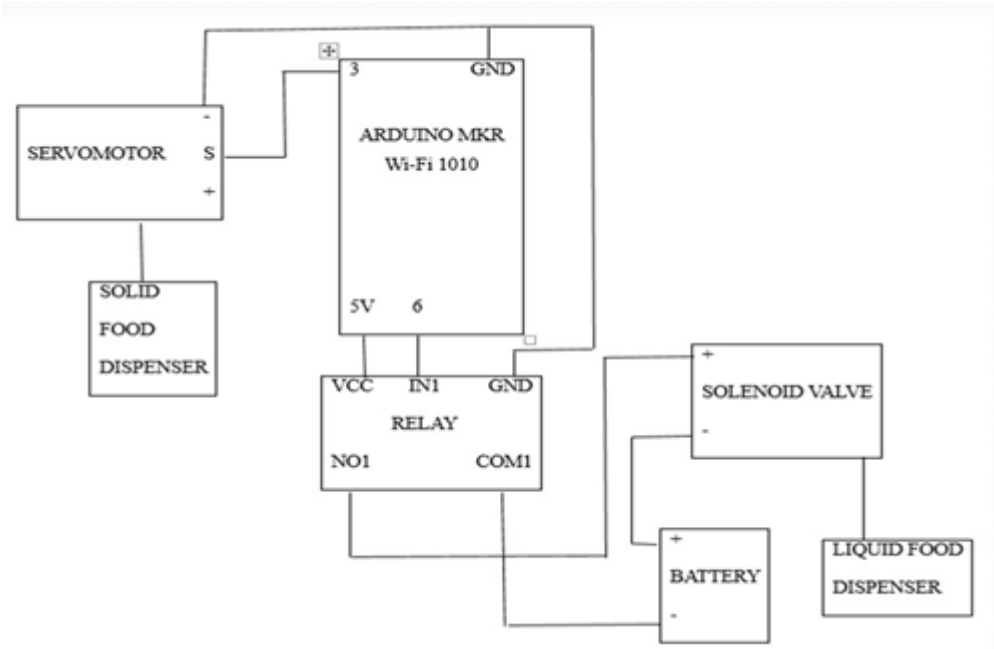

Fig4.2: Circuit diagram of proposed system

VI. CONCLUSION

Each pet owner's lifestyle is not the same, some may use their residence in the house to feed the pets, but some may not have residence or time to, this proposed pet-food feeding machine is the solution to the problem and helps taking care of their pets more conveniently. Using this machine will be different from normal way in case that the owners feed their pets by themselves, with more accurate feeding on time as we set, could be controlled from distance which normal way could not archive. This project brought together several components goal that is design an automated pet feeder using Arduino Uno. The key components of the project include a distance sensor to detect or identify the presence of the pet i.e. whether the pet is near the bowl or not and a servo motor which will be programmed to serve the food as soon as the pet comes closer. It relieves the owner from having to feed his pet multiple times a day. The proposed project senses the presence of the pet using the distance sensor and serves accordingly. The owner does not have to worry about making plans or feeding his pet because of this automated pet feeder. This automatic pet feeder serves as a helping hand as it works efficiently in the absence of the owner. This project brought together several components and ideas to an automated pet feederusing Arduino Uno.

Fig5.1:Prototype of proposed system

VII. RESULT

- We have developed a mobile application for pet owners which acts as an interface between pet owner and the pets.
- The application is available especially for people having pets.
- We have applied engineering knowledge to analyse societal problem of pet owners whenever they want to feed their pets.
- We have designed solution for it and we have analysed existing mobile apps related to pet feeder that help them to feed their pet during our project and try to overcometheir drawbacks in our project.

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